

Exploring Accessibility Barriers for Individuals Using Screen Readers on Amazon

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This study investigates the accessibility barriers which are faced by individuals who use screen readers while navigating the Amazon website. A usability test was carried out with Blind individuals who primarily use screen readers to navigate the Amazon website. The study uses qualitative methodologies, including task-based interactions and interviews, to identify important accessibility barriers such as inadequate keyboard navigation, and issues with dynamic website elements. These barriers highlight the importance of a more inclusive design strategy for screen reader users. Based on these findings, the study advises using user-centered design principles and iterative testing with screen reader users to successfully solve these difficulties. The study stresses the necessity of considering the needs of diverse users when designing an accessible online shopping experience, ensuring that dynamic and interactive information is navigable and usable by everybody, including those who rely on assistive devices such as screen readers.

CCS CONCEPTS • Accessibility → Screen Reader Usability • Human-Computer Interaction → E-commerce Accessibility

Additional Keywords and Phrases: Accessibility, E-commerce, Screen Readers, Dynamic Content, Inclusive Design

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1 INTRODUCTION

The rapid growth of e-commerce has turned most of the shopping into an online experience, enabling goods and services accessible at our fingertips [10]. However, blind and low-vision (BLV) screen reader users face various challenges such as inaccessible website navigation, information overload, dynamic content and complex page layouts that make it difficult for them to navigate through these platforms efficiently [7]. Accessibility is crucial to ensure that digital platforms are inclusive and adhere to the guidelines stated by the Web Content Accessibility Guidelines (WCAG) [2]. Despite these guidelines, e-commerce giants like Amazon fail to prioritize the accessibility of their platforms for people with disabilities [1,3].

While numerous studies have been conducted on e-commerce accessibility [8,9], this paper specifically focuses on Amazon's violations of WCAG guidelines with dynamic and interactive content such as dropdown menus, product carousels, pop-ups, and real-time updates [2]. These issues pose significant challenges for screen reader users [6]. Although

these elements are intended to improve the experience, they become barriers to accessibility. By conducting user interviews and usability testing with screen reader users, we hope to identify issues and gather qualitative data when they are interacting with these features on Amazon [4].

This study analyzes the accessibility challenges by BLV individuals when using screen readers on Amazon's website. We uncover unique issues that BLV users face when interacting with these aspects via usability testing and user interviews. Our research shows the discrepancy between WCAG compliance and screen reader users' real-world experiences, and it makes practical recommendations for increasing accessibility on Amazon's platform.

With that in mind we propose the following research questions we want to address through this research.

1.1 Research Questions

1. What specific challenges do BLV people experience when interacting with dynamic content such as pop-ups, product carousels on Amazon's website?
2. How do these challenges impact their platform navigation, usability, and task completion experiences?
3. What design changes could make Amazon's dynamic content and interactive features more accessible to screen reader users?

2 METHODS

2.1 Participants

The study involved 3 visually impaired participants. All participants rely on a screen reader. We recruited all participants with our professor, Oliver Alonzo, who helped connect us with individuals willing to participate in the study. To make sure their experiences would provide relevant data for our research, we prioritized recruiting people who frequently use screen readers for e-commerce tasks.

Table 1: Participants Demographic

Participant	Age	Gender	Visual Impairment	Screen Reader	Device
P1	76	Female	Totally Blind	JAWS	Desktop
P2	71	Female	Totally Blind	JAWS	Desktop
P3	53	Female	Totally Blind	JAWS	Laptop

2.2 Data Collection

We used a mix of questionnaires, interviews, and usability testing to gather data. Our data collection procedure is broken down into the following steps:

2.2.1 Consent Process

Before the session, we emailed the consent form to participants. To make sure they understood the goal of the study and their rights, we read the consent form out loud at the beginning of each session. Once the participants confirmed their agreement to all terms, the session proceeded to the next stage.

2.2.2 Before-Task Questions

Before every task, we asked participants if they thought the task would be easy or difficult to complete on a scale of one to five.

2.2.3 Usability Testing

We guided the participants through 3 scenarios to simulate common activity on Amazon.

1. **Using filters to find a product:** Participants used Amazon's filtering tools to look for a certain product.
2. **Putting a product in the cart:** Participants put whatever they wanted in the cart.
3. **Checkout:** To simulate making a purchase, participants walked through the checkout process.

After each scenario, participants completed the After-Scenario Questionnaire (ASQ) to evaluate their satisfaction and ease of completing the task.

2.2.4 Post-test Interview

We used interview questions to collect some qualitative data. Those questions concentrated on the difficulties they had, their thoughts on Amazon's accessibility features, and their recommendations for enhancements.

2.2.5 Data Analysis

We analyzed the data using an affinity diagram that we got from the usability testing and interviews. We organized those data into categories based on common themes. Additionally, we used inductive coding to find key themes and patterns from the participant feedback.

3 FINDINGS

We conducted the usability testing for 3 participants and all the major themes identified are summarized below:

3.1 Navigation Challenges

Participants reported different experiences while navigating through the Amazon website. Participants P1 and P2 who are in their 70s had difficulty completing the tasks and needed assistance. Whereas, participant P3 had no trouble navigating across the site because she uses keyboard shortcuts on a regular basis.

3.2 Information Overload

Even though everyone had different experiences while navigating through the site, everyone expressed that there is too much “filler” information, which makes it difficult for them to use a screen reader to go to a specific destination because it reads information line by line from left to right.

3.3 Search and Filtering Issues

All the participants expressed problems with search and filtering options. P1 and P2 preferred to manually search for products rather than using filtering options. P2 had difficulties while looking for certain price ranges, as the search results frequently displayed unrelated products. P3, on the other hand, used filters, although she noted that the search results were not always correct.

3.4 Dynamic Content

Another issue which every participant faced were the dynamic content in the form of pop ups and live recommendations that often interrupted them in doing their tasks. Even though P3 had no issues in completing tasks, she mentioned, “ *The constant pop-ups make it hard for me to stay focused on what I am doing.*”

3.5 Cart and Checkout Process

Participants P1 and P2 expressed difficulty with adding items to the cart as they were unsure of the items they added to the cart and had to double check the items they were purchasing at the checkout process because there were too many pop-ups there. P3 who is very familiar with JAWS did not have this problem although she struggled with the countdown timer on timed discounts stressful as it hurried her through the checkout process.

3.6 Accessibility Assistance and Support

P1 and P2 mentioned that they use Amazon’s accessibility line in case they are stuck. Upon conducting user interviews, they mentioned that the experience is not always positive and varies from time to time. P2 also recalled Amazon Access, stating, “*There used to be Amazon Access, and it was great but now it’s gone.*” P3 mentioned that they don’t require any assistance, and they use keyboard shortcuts and rarely if they need help, they take help of their sighted husband.

4 DISCUSSION

4.1 Process and Findings

Our main goal in this study was to identify accessibility issues on shopping websites, focusing on Amazon as it is considered the largest e-commerce platform in the U.S.

Given the limited time and the number of participants meeting our criteria, we used qualitative methods such as user interviews and usability testing. These methods involved interacting with Amazon's iterative design based on a set of research questions.

We recruited a total of three users through our professor’s connections. Our research aimed to address questions related to dynamic and interactive elements. To gather additional insights, participants completed pre and post task questionnaires to capture their feelings and experiences before and after each task. Observational notes were also taken. The data collected from the interviews was analyzed, and an affinity diagram was used to gain further insights. The findings revealed that all users encountered issues related to navigation, search and filtering functionality, the checkout process, and accessibility support.

4.2 Results Evaluation

Some of the participants' issues varied depending on their familiarity with using shortcuts in a screen reader. One participant was unaware of shortcuts, another had limited knowledge but hadn’t fully mastered them, while the third participant was highly proficient in using shortcuts. As we only interviewed participants of ages 50+ that might be an accessibility challenge for older users of screen readers as they might not keep up with new technologies. Two out of the three participants used Amazon’s assistance reporting different experiences while the proficient shortcuts user rarely used assistance.

4.3 Results and Literature Relation

The results evaluation aligns with the literature cited in the introduction in several ways by confirming some of the accessibility challenges identified in prior studies that were mentioned. The participants' experience familiarity with screen reader shortcuts notes a major issue for older users adapting to new technologies and updates, reflecting a challenge of inaccessible navigation and makes users rely on user experience mentioned in the literature. The challenges faced by P1 and P2 with dynamic content, such as pop-ups and real-time updates, and how all of the three participants had navigation issues with price sliders directly line up with the barriers described in the introduction as violations of WCAG guidelines. The mixed experiences with Amazon's accessibility support highlighted the gap between meeting accessibility standards that are usable for BLV users. This results in the need for more improved and equitable designs and accessibility features on e-commerce platforms as outlined in the introduction. Therefore, the findings corroborate the literature's emphasis on addressing these problems to create a truly accessible e-commerce environment for users.

4.4 Design implications

Our findings highlighted several design improvements to enhance accessibility for screen reader users.

4.4.1 Simplified Navigation and content.

Use clear heading structures, logical grouping and skip links to reduce cognitive strain. Reduce non-essential information to improve screen reader navigation.

4.4.2 Dynamic Content Controls

Reduce disruptions from pop-ups and live updates by giving users the choice to disable these features and only use ARIA live zones when necessary.

4.4.3 Accessible filtering and checkout

Make sure the filtering tools are completely keyboard navigable and provide real-time feedback on cart activities. Introduce user-controlled timers for features like discounts to reduce stress.

4.4.4 Enhanced Accessibility Support

Provide support services for users by providing consistent, informed help to screen reader users.

4.4.5 Onboarding and education

Provide accessible tutorials on screen reader shortcuts and navigation tactics to help screen reader users.

4.5 Limitations of the Current Study

Our study had several limitations that prevented us from delving deeper into the research and extracting more comprehensive results to make significant adjustments or broader impacts. These limitations primarily included the sample size, participant diversity, and time constraints.

We worked with three participants, all of whom had complete blindness, whereas our goal was to study both blind and low-vision users. Additionally, the limited timeframe restricted our ability to recruit more participants, resulting in a smaller, less diverse sample with limited variation in age and background.

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